Somatic read simulators: how to build a realistic gold standard using data from well-annotated cancer databases

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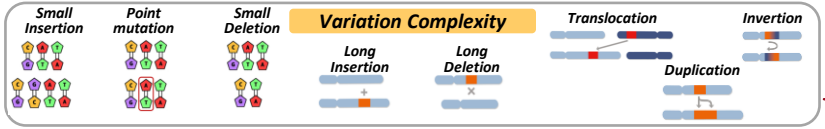
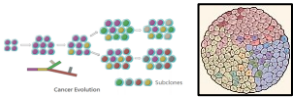
Robust & flexible simulation framework of tumoral variability



Inter patient variation
- Histology
- Grade

Intra patient variation
- Primary
- Metastatic

Intra sample variation
- Abnormal number of chromosomes
- Sample contamination
- Variants zigosity
- Presence of subclones



	Svengine	BAMsurgeon	VarSim	Xome Blender	SCNVsim	Pysim SV	THap Mix
Germinal Variants	✓	✗	✓	✗	✓	✓	✗
Somatic Variants	✓	✓	✓	✓	✓	✓	✓
Variants number	✓	✓	✓	✓	✗	✓	✗
Variants length	✓	✓	✓	✗	✗	✓	✗
Variants position	✓	✓	✗	✗	✗	✗	✓
Ploidy	✗	✗	✗	✗	✓	✓	✓
Purity	✗	✓	✓	✓	✓	✓	✓
Zigosity	✓	✗	✓	✗	✓	✓	✗
Clonality	✓	✗	✗	✓	✓	✓	✓

Tumor-specific & data-driven presets

COSMIC Catalogue Of Somatic Mutations In Cancer

The Cancer Genome Atlas

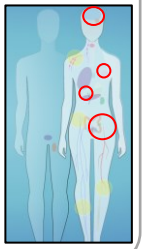
CCLE Cancer Cell Line Encyclopedia

International Cancer Genome Consortium

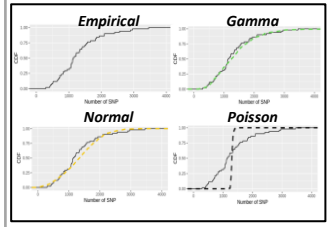
Learning

- Numerosity
- Lengths
- Positions

of SNP, INS, DEL, DUP, DELINS in



Fitting



AIC BIC KS AD CvM

SVengine + PYsim-SV

	SNP	INS	DEL	...
Breast cancer				
Lung cancer				
Colon cancer				
...				

Realistic & reliable somatic sample simulation

What's next?

Validation of somatic variant calling pipelines